

Project1 for Cloud Computing Fall 2026

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For this project, you are tasked to write Go program that calls a programming API (Application Programming Interface) and deliver to consumers as Docker images. Go programming language, API's, and Docker are used frequently in cloud software development environments.

The user is interested in searching for news on a topic of their choice. They will enter few words as their search term and you need to collect and present news related to that search topic using a website such as newsapi.org. A command line interface (CLI) will be sufficient for the user interface.

The user will also enter two numbers, one defining how many days of news to be searched. For example, entering 2 will search for news today and yesterday. Also, an upper limit will be entered, such as 20, and this will limit the number of news to be delivered.

A database of choice will be implemented to buffer the results. If a new user uses the same search query, the existing results will be read from the database instead of reaching the API. However if the user increased the time period or the news count limit, new results will be added to the database and the API needs to be used.

AI use is permitted, however all prompts sent to the AI needs to be reported in the documentation including the specific prompts used. Also if used, AI name and version need to be reported.

The program has to use go routines, concurrency, and channels. It should allow as many calls from different users at the same time and this feature should use the concurrency and channels of the same program. You can specify a specific way of calling your code multiple times at once and describe in your documentation.

A Docker image, docker file, and source code need to be submitted separately. The latter two should be able to produce the submitted Docker image.

The size of the docker image should be minimal. Your submission will be compared to those of other students and your project score will be adjusted accordingly.

Furthermore, the latency of your implementation and number of simultaneous requests may also affect your score.

Teams of one or two people are acceptable. The submission needs to be accompanied by a report that describes how to compile and run your code. Include input testcase files you may have used to ensure that your code is working properly. The report should discuss the following:

1. Software architecture used in the code and why these decisions were made.
2. Explain the tests used to validate the work and what was discovered.
3. Citations to previous work, such as papers or online work and discussions of how your work may be different. Please use an IEEE or ACM citation format either for a journal or conference paper, be consistent once you have picked your format.
4. All prompts sent to AI and AI name & version used in a way that your work could be replicated by reusing these.

All source code, instructions to run the code, all test inputs and outputs should also be submitted.

If you post online about this work, you should cite the course and this document.

Including something surprising and better than other teams may unlock a 10% bonus.